

Existence, Uniqueness, Blow-Up, and Potentials

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Definition: Maximal solution

A **maximal solution** is a solution whose interval of definition is not properly contained in the interval of definition of any extension satisfying the same initial-value problem.

Definition: Global forward solution

A solution through time t_0 is **global forward in time** if it is defined for every $t \geq t_0$.

Definition: Finite-time blow-up

Let $x : (\alpha, t_*) \rightarrow \mathbb{R}$ be a maximal solution, where $t_0 \in (\alpha, t_*)$ and $t_* < \infty$. The solution has **forward finite-time blow-up at t_*** if

$$\lim_{t \uparrow t_*} |x(t)| = \infty.$$

Definition: Locally Lipschitz map

Let $U \subseteq \mathbb{R}^n$ be open. A map $F : U \rightarrow \mathbb{R}^m$ is **locally Lipschitz** if, for every $p \in U$, there are a neighborhood $N \subseteq U$ of p and a constant $L \geq 0$ such that

$$\|F(x) - F(y)\| \leq L\|x - y\|$$

for all $x, y \in N$.

Theorem: Local existence and uniqueness

Consider

$$\dot{x} = f(t, x), \quad x(t_0) = x_0.$$

If f is continuous on a closed rectangle whose interior contains (t_0, x_0) , and if there is an $L \geq 0$ such that

$$|f(t, x) - f(t, y)| \leq L|x - y|$$

whenever (t, x) and (t, y) lie in that rectangle, then there is a unique local solution.

Theorem: Local existence and uniqueness (continued)

Proof sketch. On a smaller closed rectangle, let $|f(t, x)| \leq M$ and define the Picard iterates

$$x_0(t) \equiv x_0, \quad x_{n+1}(t) = x_0 + \int_{t_0}^t f(s, x_n(s)) ds.$$

On a sufficiently short interval, all iterates remain in the rectangle and induction gives

$$|x_{n+1}(t) - x_n(t)| \leq \frac{ML^n |t - t_0|^{n+1}}{(n+1)!}.$$

The series of successive differences therefore converges uniformly to a solution of the integral equation. If x and y are two solutions with the same initial value, then, for $t \geq t_0$,

$$|x(t) - y(t)| \leq L \int_{t_0}^t |x(s) - y(s)| ds.$$

Grönwall's inequality gives $x(t) = y(t)$ for $t \geq t_0$. Applying the same argument backward from t_0 proves uniqueness on both sides. $\square$

Corollary: Autonomous local existence and uniqueness

Let $I \subseteq \mathbb{R}$ be open, let $f \in C^1(I)$, and let $x_0 \in I$. Then

$$\dot{x} = f(x), \quad x(0) = x_0,$$

has a unique local solution.

Proof. On every compact interval contained in I , the derivative f' is bounded. The mean value theorem therefore makes f locally Lipschitz, so the local existence-and-uniqueness theorem applies. $\square$

Question: How can one initial condition have infinitely many futures?

On the state space $x \geq 0$, solve

$$\dot{x} = x^{1/3}, \quad x(0) = 0,$$

and exhibit infinitely many solutions.

Solution. The constant function $x(t) = 0$ is a solution. For every waiting time $a \geq 0$, define

$$x_a(t) = \begin{cases} 0, & t \leq a, \\ \left(\frac{2(t-a)}{3}\right)^{3/2}, & t \geq a. \end{cases}$$

For $t > a$,

$$\dot{x}_a(t) = \left(\frac{2(t-a)}{3}\right)^{1/2} = x_a(t)^{1/3}.$$

At $t = a$, both $\dot{x}_a(a)$ and $x_a(a)^{1/3}$ equal 0, so x_a is differentiable and satisfies the equation there as well. Every x_a also satisfies $x_a(0) = 0$. Distinct waiting times give distinct solutions. The vector field is continuous but is not locally Lipschitz at 0.

Question: Does a smooth vector field guarantee global existence?

Solve

$$\dot{x} = 1 + x^2, \quad x(0) = 0,$$

and find its maximal interval.

Solution. The vector field is smooth, so the local solution is unique. Separation gives

$$\int_0^{x(t)} \frac{du}{1+u^2} = t, \quad \arctan x(t) = t.$$

Therefore

$$x(t) = \tan t.$$

Its maximal interval is

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

In particular,

$$\lim_{t \uparrow \pi/2} x(t) = +\infty,$$

so the maximal solution has finite-time blow-up at $t_* = \pi/2$ and is not global forward in time.

Theorem: No nonconstant periodic solution on a line

Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be continuous. Every periodic solution of

$$\dot{x} = f(x)$$

is constant.

Proof. Let F be an antiderivative of f , and suppose $x(t+T) = x(t)$ for some $T > 0$. Then

$$\begin{aligned} 0 &= F(x(t+T)) - F(x(t)) \\ &= \int_t^{t+T} F'(x(s)) \dot{x}(s) ds \\ &= \int_t^{t+T} f(x(s)) \dot{x}(s) ds \\ &= \int_t^{t+T} \dot{x}(s)^2 ds. \end{aligned}$$

The integrand is continuous and nonnegative, so it is identically zero. Hence $\dot{x}(t) = 0$ for every t , and x is constant. $\square$

Definition: Potential for a scalar autonomous equation

Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be continuous. A **potential** for

$$\dot{x} = f(x)$$

is a differentiable function $V : I \rightarrow \mathbb{R}$ satisfying

$$f(x) = -V'(x).$$

Equivalently,

$$V(x) = - \int_{x_{\text{ref}}}^x f(s) ds + C,$$

where $x_{\text{ref}} \in I$ and $C \in \mathbb{R}$ are arbitrary.

Observation: Equilibria are critical points

For $\dot{x} = -V'(x)$,

$$x^* \text{ is an equilibrium} \iff V'(x^*) = 0.$$

Theorem: Potential descent identity

Let $I \subseteq \mathbb{R}$ be an interval, let $V : I \rightarrow \mathbb{R}$ be a potential for $\dot{x} = f(x)$, and let $x : J \rightarrow I$ be a solution. For every $t \in J$,

$$\frac{d}{dt}V(x(t)) = -\left(f(x(t))\right)^2 = -\left(V'(x(t))\right)^2 \leq 0.$$

Equality holds at time t exactly when $x(t)$ is an equilibrium.

Proof. By the chain rule and $\dot{x} = f(x) = -V'(x)$,

$$\frac{d}{dt}V(x(t)) = V'(x(t))\dot{x}(t) = V'(x(t))(-V'(x(t))) = -\left(V'(x(t))\right)^2.$$

The square vanishes exactly when $V'(x(t)) = 0$, equivalently when $f(x(t)) = 0$. □

Definition: Potential well

A strict local minimum of a potential is a **well**.

Definition: Potential barrier

Let $x_- < x_b < x_+$, where x_- and x_+ are wells of a scalar potential V . If x_b is a local maximum of V , then x_b is a **potential barrier** between those two wells.

Definition: Bistability

A system on a specified state space is **bistable** if it has exactly two distinct asymptotically stable equilibria in that state space.

Corollary: Nondegenerate extrema and stability

Let $I \subseteq \mathbb{R}$ be an open interval, let $V \in C^2(I)$ be a potential for $\dot{x} = -V'(x)$, and let $x^* \in I$ satisfy $V'(x^*) = 0$.

1. If $V''(x^*) > 0$, then x^* is locally asymptotically stable.
2. If $V''(x^*) < 0$, then x^* is unstable.

Proof. Since $f = -V'$,

$$f'(x^*) = -V''(x^*).$$

The two conclusions follow from the scalar linear stability test. □

Question: How does a double-well potential encode bistability?

For

$$\dot{x} = x - x^3,$$

find a potential, classify every equilibrium, and determine the two basins of attraction.

Solution. Since $V'(x) = -f(x) = -x + x^3$, integration gives

$$V(x) = -\frac{x^2}{2} + \frac{x^4}{4}.$$

The equilibria satisfy

$$x - x^3 = x(1 - x^2) = 0,$$

so they are -1 , 0 , and 1 . Also,

$$V''(x) = -1 + 3x^2,$$

and hence

$$V''(0) = -1 < 0, \quad V''(\pm 1) = 2 > 0.$$

Thus 0 is an unstable barrier, while -1 and 1 are asymptotically stable wells. The phase-line sign table is

x	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
$x - x^3$	+	-	+	-

and therefore

$$\mathcal{B}(-1) = (-\infty, 0), \quad \mathcal{B}(1) = (0, \infty).$$

The initial condition $x_0 = 0$ gives the constant solution at the barrier. Thus the system is bistable on $\mathbb{R}$.